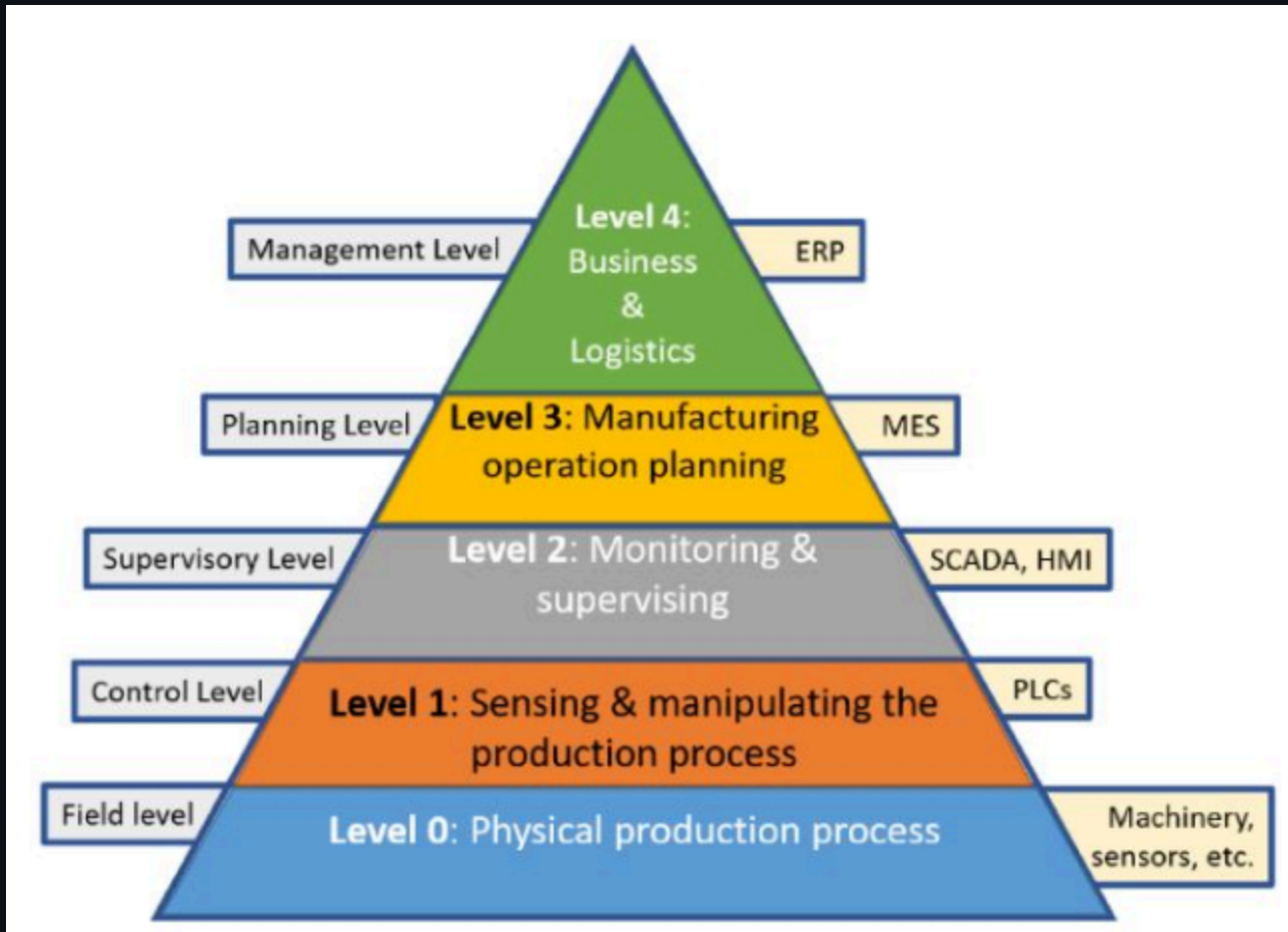


Production Supporting Systems in Factories

Smart Sensors and Information Systems

อุปกรณ์ตรวจวัดอัจฉริยะและระบบสารสนเทศ

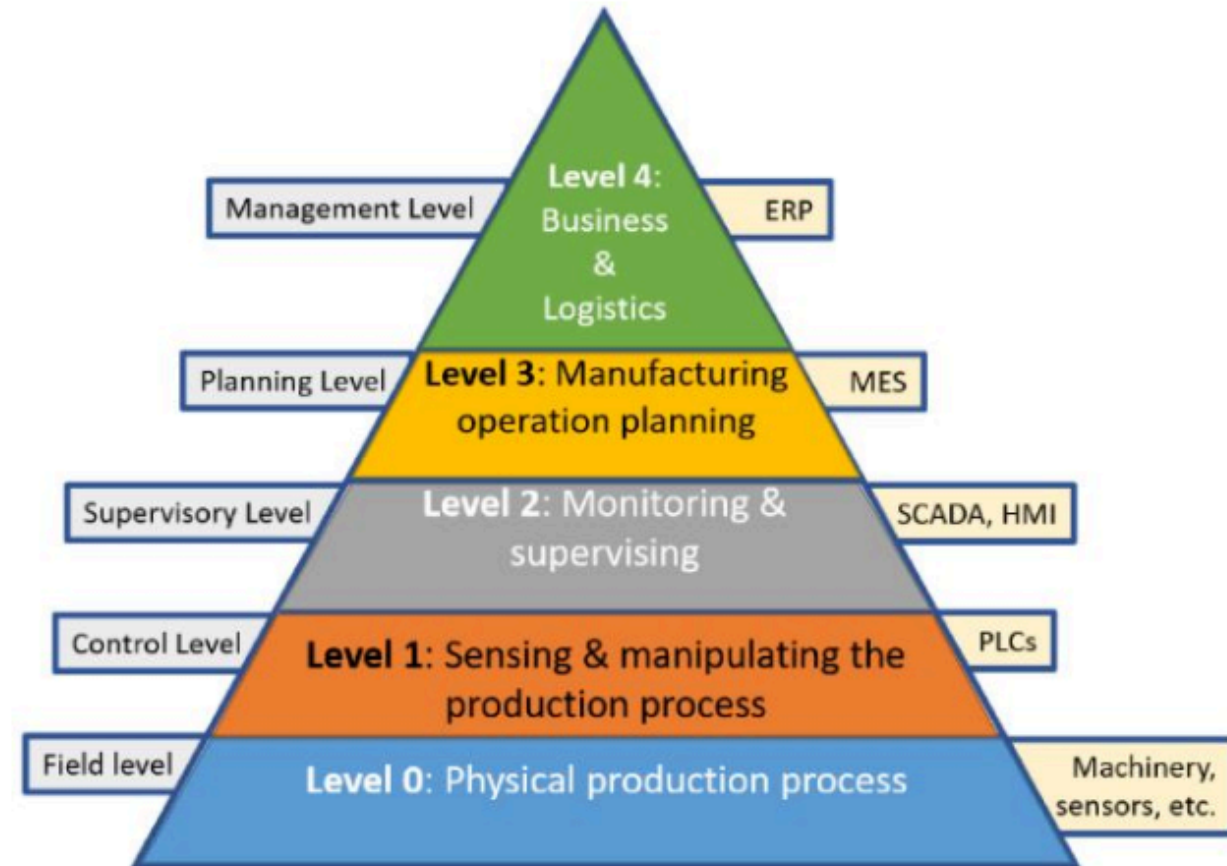


ISA-95 Standard

- International standard for the integration of enterprise and control systems.
- Defines the hierarchy of systems in manufacturing operations.

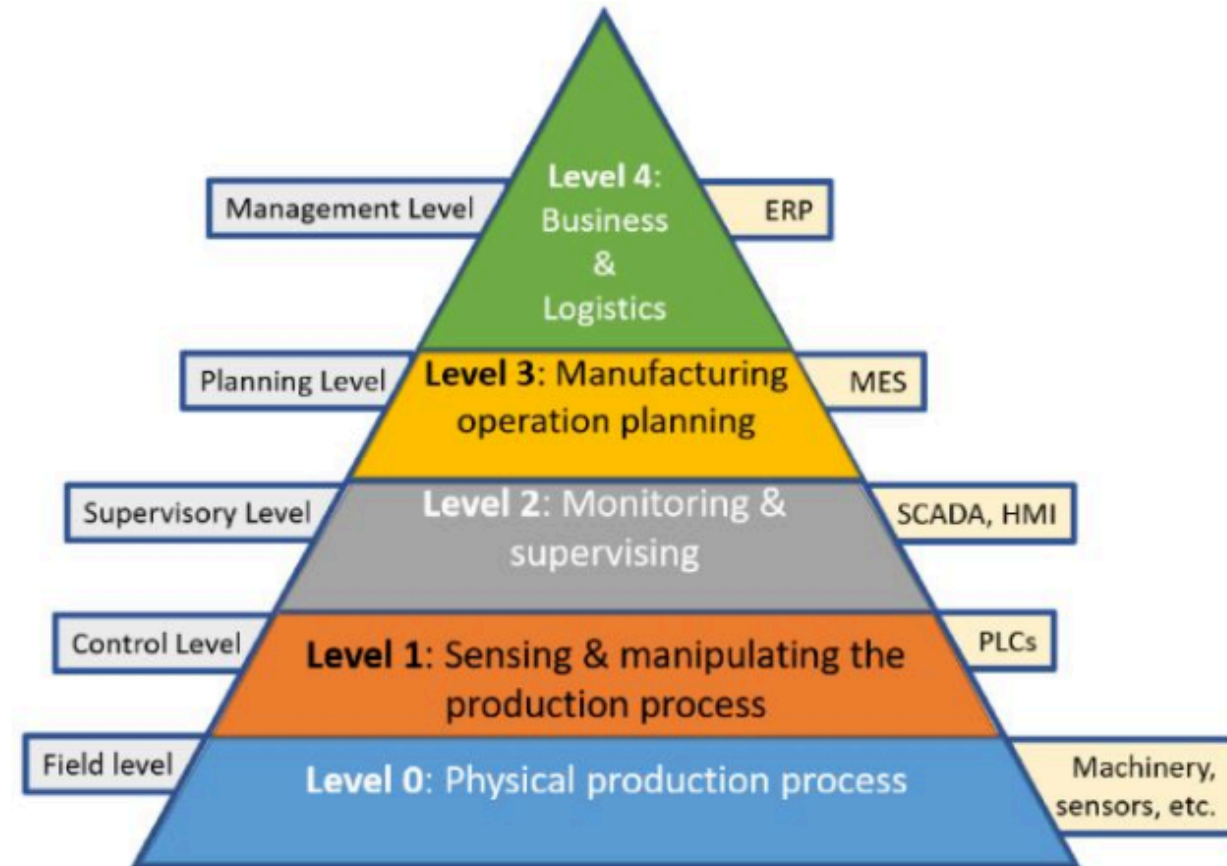
Level 0

- **Production Process**
- The actual physical and chemical processes on the plant floor.



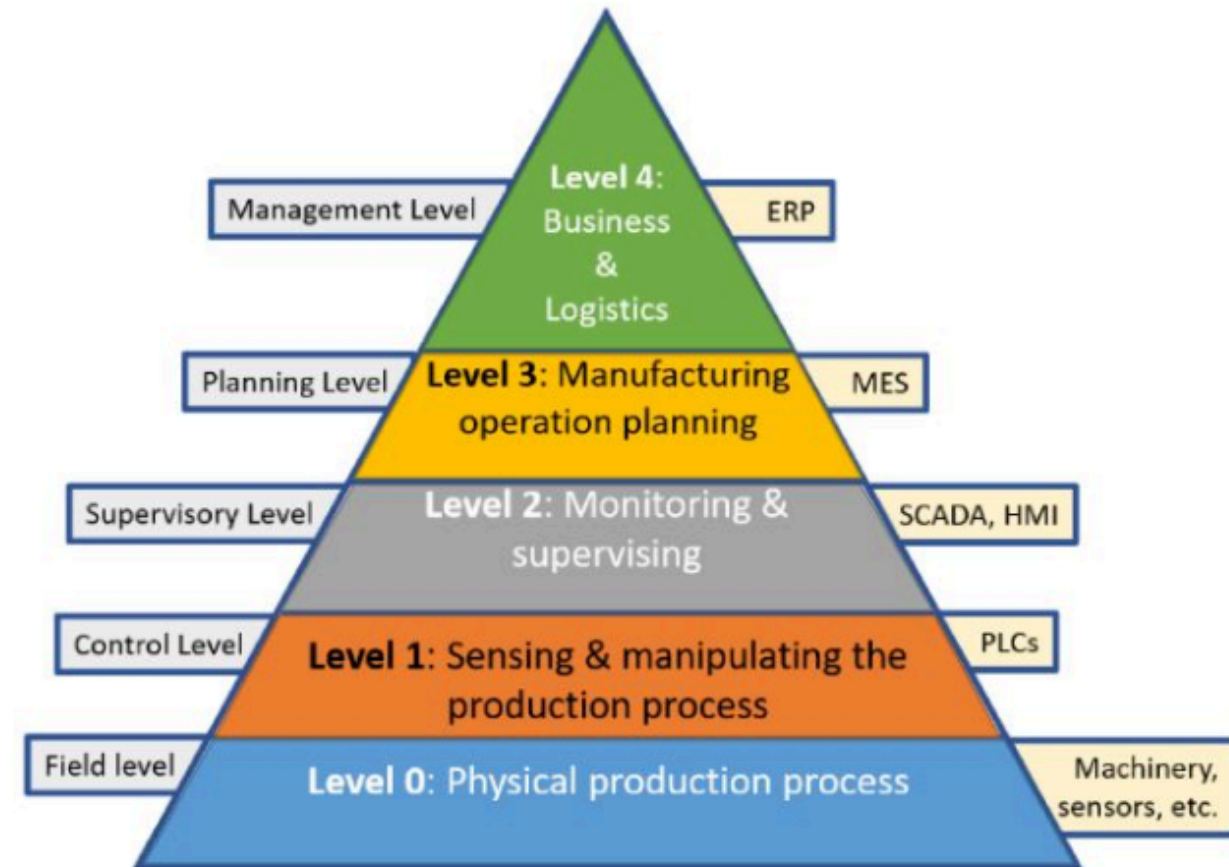
Level 1

- **Sensing and Manipulation**
- Sensors and actuators that interact directly with the production process.



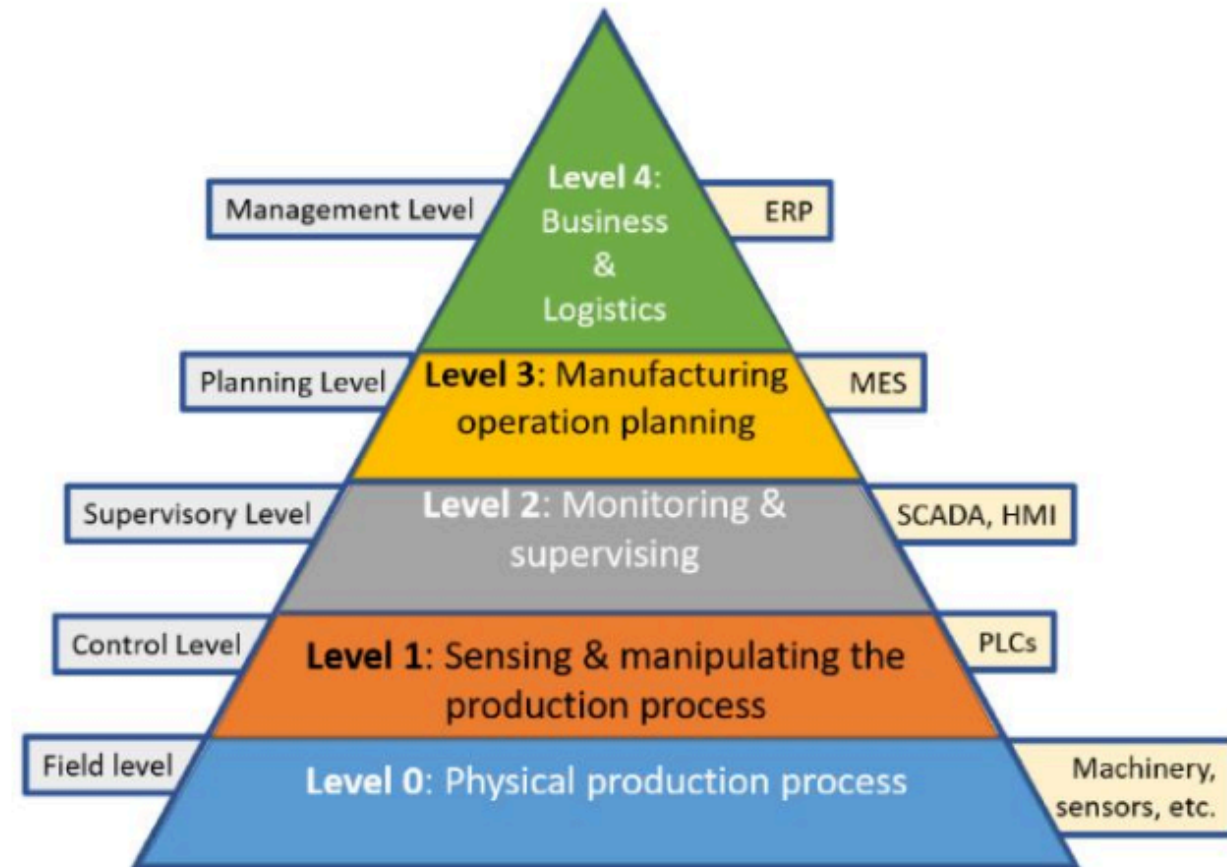
Levels 2

- **Monitoring and Supervision**
- Systems like SCADA and HMI that monitor and control the process in real-time.



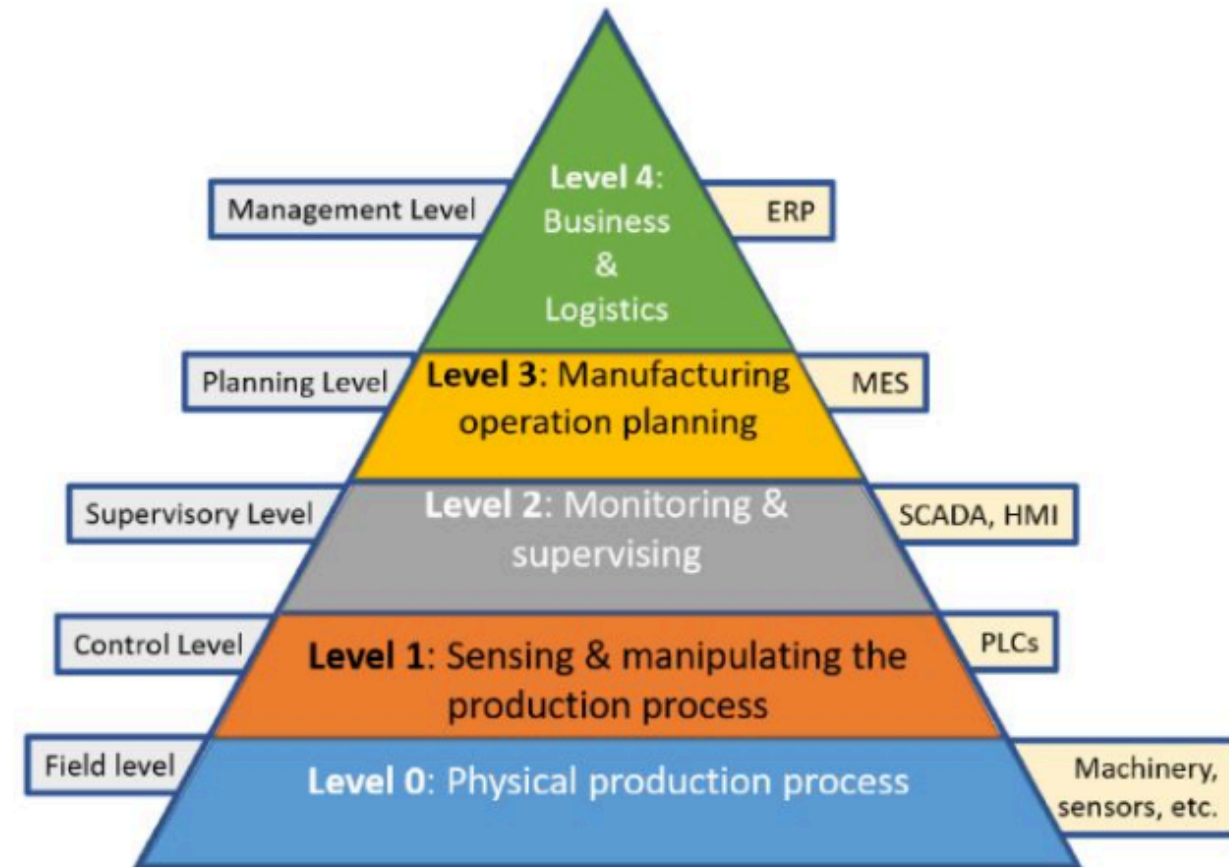
Levels 3

- **Manufacturing Operations Management**
- Manages day-to-day production, including scheduling, quality, and inventory. MES (Manufacturing Execution System) is a key component here.



Levels 4

- **Business Planning and Logistics**
- Handles enterprise-level functions like production planning, order processing, and supply chain management, typically managed by ERP systems.

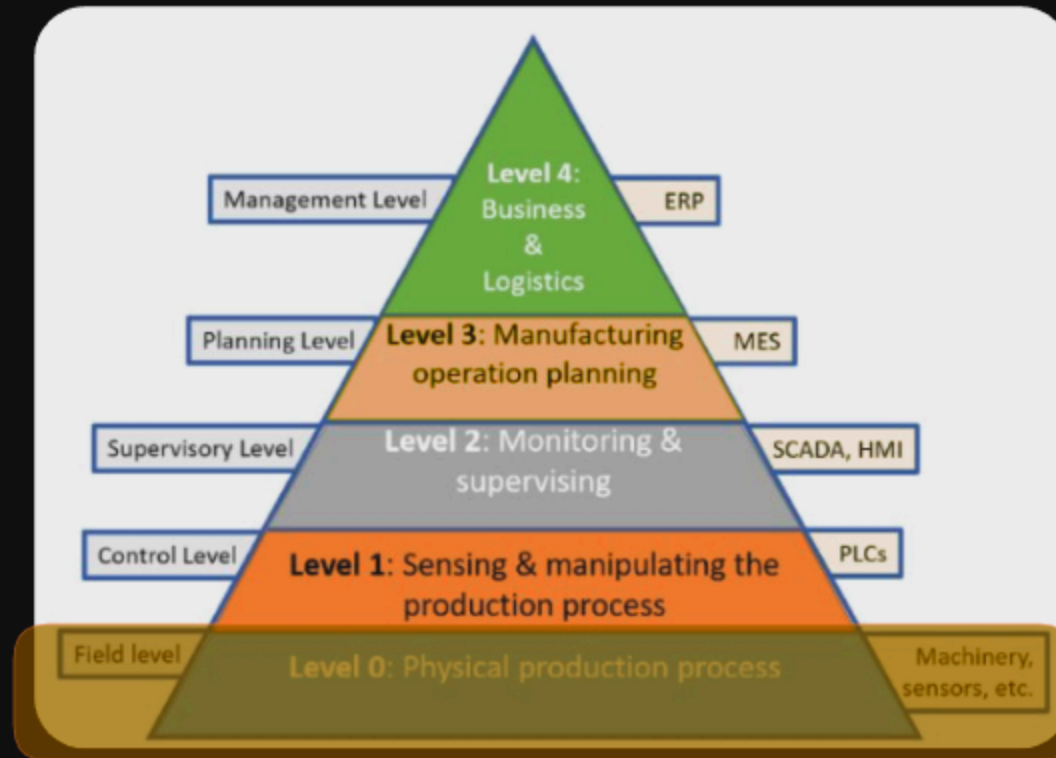


Note

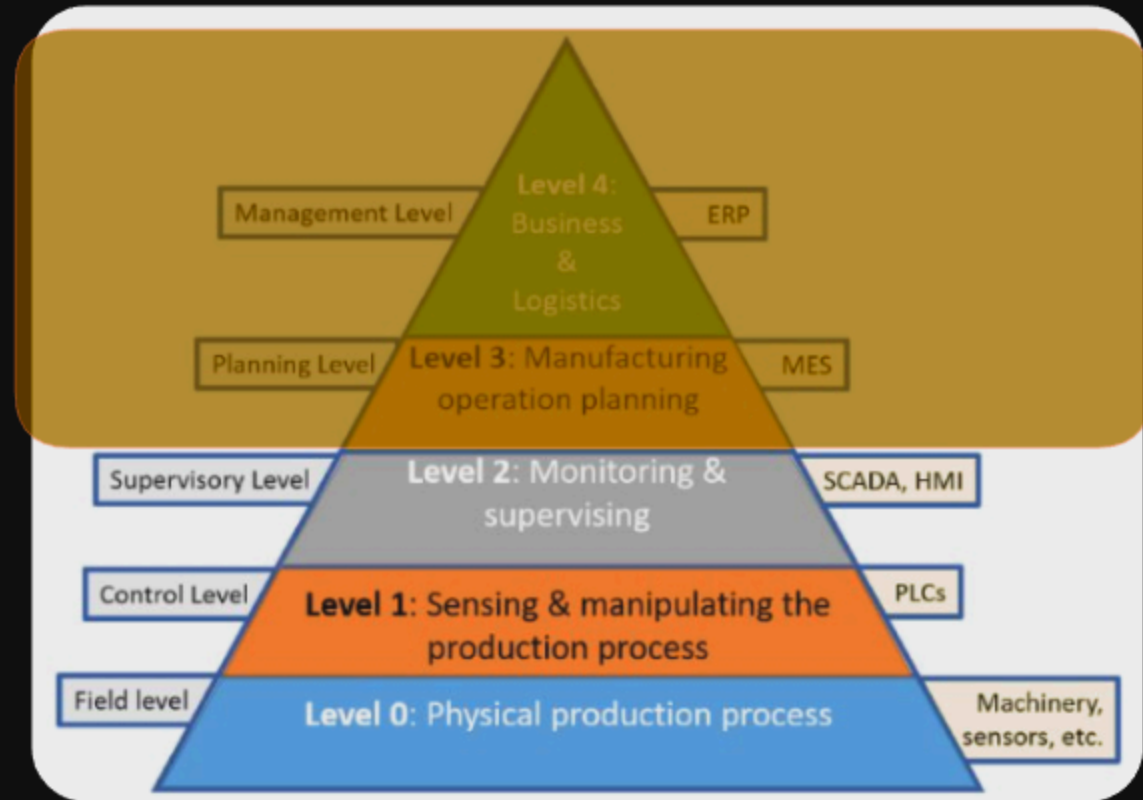
- This diagram is rather conceptual and may not represent all real-world scenarios.
- In practice, systems can overlap and interact in complex ways.

What are we doing in this course?

- We will use sensors and software that are related to these levels.
- I only pick the options that are accessible and affordable.
 - Open-source softwares
 - Low-cost sensors and devices
- But they are still relevant to real-world applications.



ERPNext



Week	Class	Topic
22 Dec	1, 2	ERPNext (Intro, Purchase, Sales)
29 Dec		<i>Holiday</i>
5 Jan	3, 4	ERPNext (Inventory, Manufacturing)
19 Jan	5, 6	Node-Red (Basic, MQTT)
26 Jan		<i>Reading Week</i>
2 Feb		<i>Midterm Week</i>
9 Feb		Project

Let's go...